Reinforcement Learning

Dynamic Programming

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Policies and Value Functions

- A policy π is better than π' if $v_\pi(s) \geq v_{\pi'}(s)$ for all $s \in \mathcal{S}$.
- Optimal policy: better than or equal to all other policies
- Optimal state-value function:

$$v_*(s) := \max_\pi v_\pi(s) \quad \text{for all } s \in \mathcal{S}$$

- Optimal action-value function:

$$q_*(s,a) := \max_\pi q_\pi(s,a) \quad \text{for all } s \in \mathcal{S}, a \in \mathcal{A}(s)$$

- can write q_* in terms of v_*

$$q_*(s,a) = \mathbb{E}\left[R_{t+1} + \gamma v_*(S_{t+1}) \mid S_t = s, A_t = a\right]$$

Policies and Value Functions

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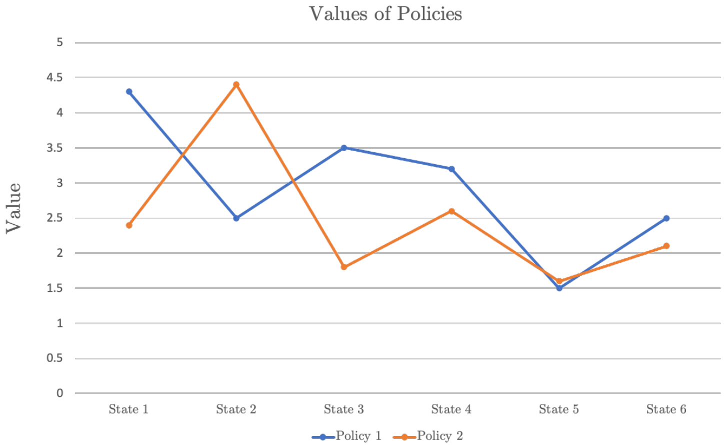

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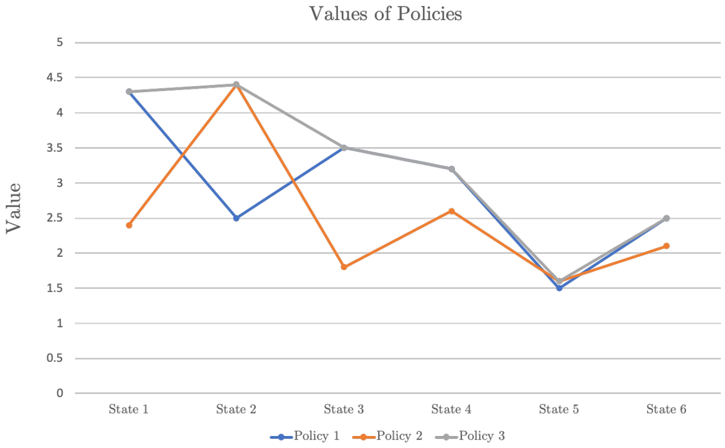

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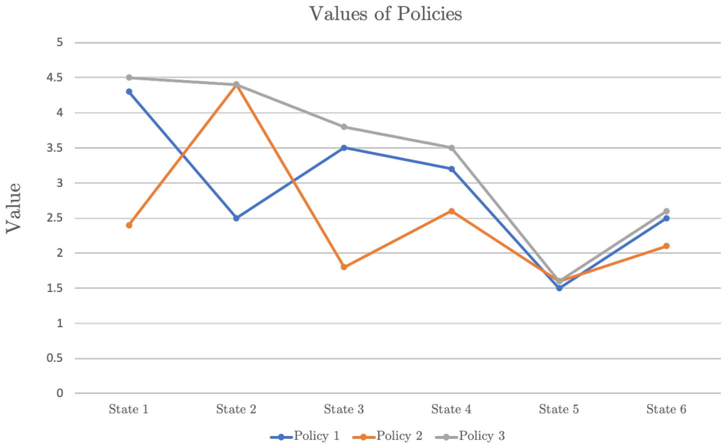

Dynamic Programming (DP)

- DP provides an essential foundation for the understanding of the methods for RL.
- Most modern RL methods essentially attempt to achieve the same effect as DP.
  - with less computation and without assuming a perfect model of the environment

Policy Evaluation & Policy Improvement

Policy Evaluation

- Iterative computation of the value function for a given policy
  - Given MDP and policy π, compute v_π
- Also called as prediction problem

Policy Improvement

- Computation of an improved policy given the value function
  - Given MDP and value function v_π, find improved policy π'
- Also known as control

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Policy Evaluation Update Rule

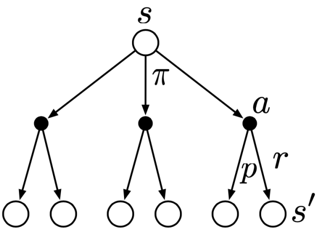
- Recall the Bellman equation: for all $s \in \mathcal{S}$

$$
\begin{aligned}
v_\pi(s) &:= \mathbb{E}_\pi \left[G_t \mid S_t = s \right] \\
&= \mathbb{E}_\pi \left[R_{t+1} + \gamma G_{t+1} \mid S_t = s \right] \\
&= \sum_a \pi(a|s) \sum_{s'} \sum_r p(s', r|s, a) \left[r + \gamma v_\pi(s') \right]
\end{aligned}
$$

- Consider a sequence of value functions $v_0, v_1, v_2, \ldots$: approximates of value. function v_π under policy π
- Successive approximation using the Bellman equation for all $s \in \mathcal{S}$,

$$
v_{k+1}(s) \leftarrow \sum_a \pi(a|s) \sum_{s'} \sum_r p(s', r|s, a) \left[r + \gamma v_k(s') \right]
$$

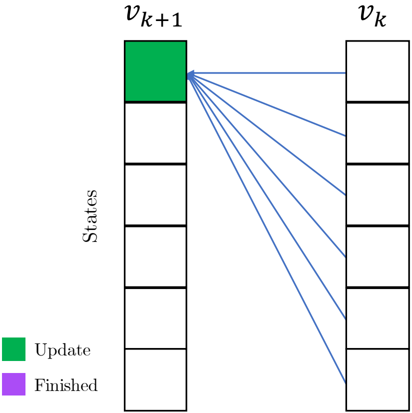
Update using Bellman Backups



$$v_{k+1}(s) \leftarrow \sum_a \pi(a|s) \sum_{s'} \sum_r p(s', r|s, a) \left[r + \gamma v_k(s')\right]$$

- Update $v_{k+1}(s)$ using the value information of leaf states $v_k(s')$

Value Function Updates with Two Arrays



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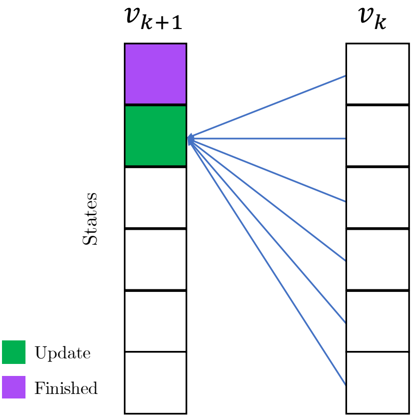

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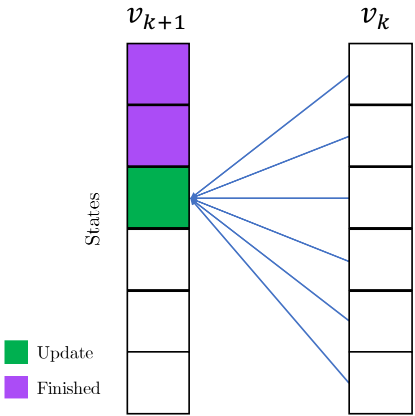

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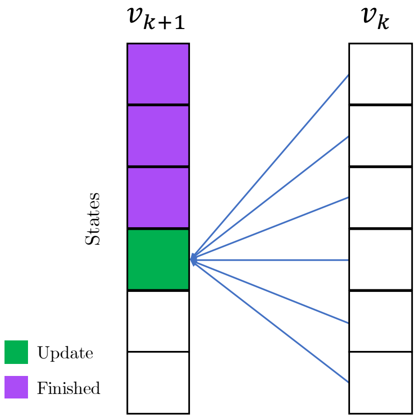

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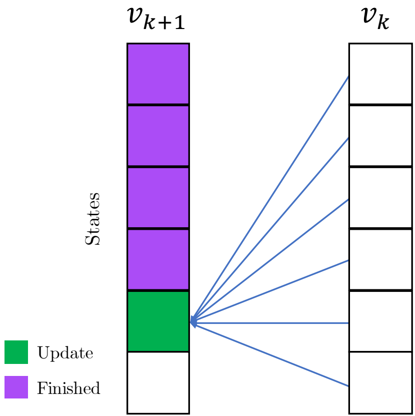

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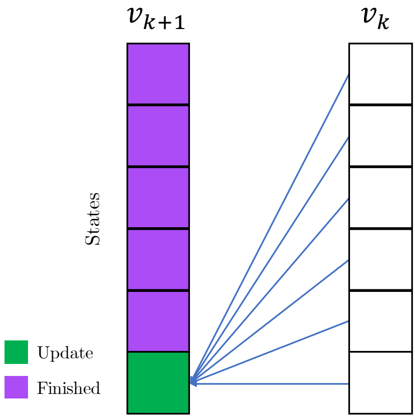

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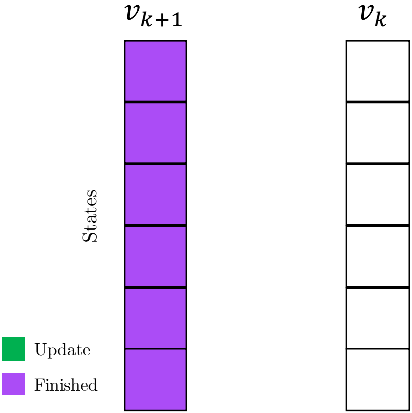
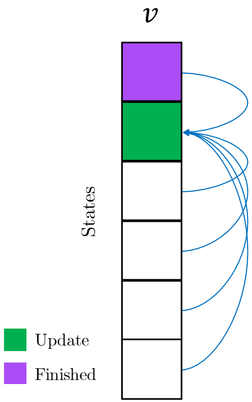


Value Function “In-place” Updates



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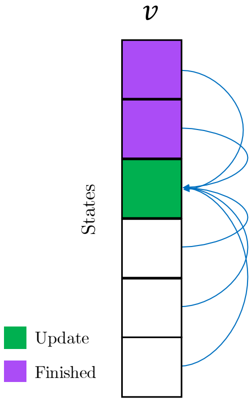


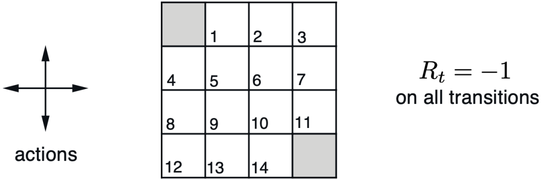
Iterative Policy Evaluation

Iterative Policy Evaluation, for estimating $V \approx v_\pi$

Input π, the policy to be evaluated
Algorithm parameter: a small threshold $\theta > 0$ determining accuracy of estimation
Initialize $V(s)$, for all $s \in \mathcal{S}^+$, arbitrarily except that $V(\mathit{terminal}) = 0$

Loop:
  $\Delta \leftarrow 0$
  Loop for each $s \in \mathcal{S}$:
    $v \leftarrow V(s)$
    $V(s) \leftarrow \sum_a \pi(a|s) \sum_{s',r} p(s', r | s, a) \left[r + \gamma V(s') \right]$
    $\Delta \leftarrow \max(\Delta, |v - V(s)|)$
until $\Delta < \theta$

Example: 4×4 Gridworld



- Non-terminal states $\mathcal{S} = \{1, 2, \ldots, 14\}$ and *shaded* terminal states
- Actions $\mathcal{A} = \{\texttt{up}, \texttt{down}, \texttt{right}, \texttt{left}\}$
- *Deterministic* state transition & reward model $p(s', r|s, a)$
  - e.g., $p(6, -1|5, \texttt{right}) = 1$, $p(7, -1|7, \texttt{right}) = 1$, $p(10, r|5, \texttt{right}) = 0$ for all $r \in \mathcal{R}$
- What do you achieve if you are maximizing sum of rewards?

Policy Evaluation in Gridworld

v_π for random policy

$k = 0$

0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0

$k = 1$

$k = 2$

- Two array updates using $v_{k=1}$ and v_k

$$v_{k+1}(s) \leftarrow \sum_a \pi(a|s) \sum_{s',r} p(s',r|s,a)\left[r + \gamma v_k(s')\right]$$

- Uniform random policy $\pi(a|s) = 0.25$ for all $a \in \mathcal{A} = \{\texttt{up}, \texttt{down}, \texttt{right}, \texttt{left}\}$ and for all non-terminal states $s \in \mathcal{S}$
- No transitions from terminal states
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Policy Evaluation in Gridworld

v_π for random policy

$k = 0$

0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0

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Policy Evaluation in Gridworld

v_π for random policy

$k = 0$

0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0

$k = 1$

		-1.0	

$k = 2$

- Two array updates using $v_{k=1}$ and v_k

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0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0

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Policy Evaluation in Gridworld

v_π for random policy

$k = 0$

0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0

$k = 1$

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$k = 2$

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$k = 0$

0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0

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$k = 0$

0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0

$k = 1$

0.0	-1.0	-1.0	

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$k = 0$

0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0

$k = 1$

0.0	-1.0	-1.0	

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v_π for random policy

$k = 0$

0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0

$k = 1$

0.0	-1.0	-1.0	-1.0
-1.0	-1.0	-1.0	-1.0
-1.0	-1.0	-1.0	-1.0
-1.0	-1.0	-1.0	0.0

$k = 2$

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v_π for random policy

$k = 0$

0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0

$k = 1$

0.0	-1.0	-1.0	-1.0
-1.0	-1.0	-1.0	-1.0
-1.0	-1.0	-1.0	-1.0
-1.0	-1.0	-1.0	0.0

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Policy Evaluation in Gridworld

v_π for random policy

$k = 0$

0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0

$k = 1$

0.0	-1.0	-1.0	-1.0
-1.0	-1.0	-1.0	-1.0
-1.0	-1.0	-1.0	-1.0
-1.0	-1.0	-1.0	0.0

$k = 2$

		-2.0	

- Two array updates using $v_{k=1}$ and v_k

$$v_{k+1}(s) \leftarrow \sum_a \pi(a|s) \sum_{s',r} p(s', r|s, a) \left[r + \gamma v_k(s')\right]$$

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v_π for random policy

$k = 0$

0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0

$k = 1$

0.0	-1.0	-1.0	-1.0
-1.0	-1.0	-1.0	-1.0
-1.0	-1.0	-1.0	-1.0
-1.0	-1.0	-1.0	0.0

$k = 2$

		-2.0	

- Two array updates using $v_{k=1}$ and v_k

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Policy Evaluation in Gridworld

v_π for random policy

$k = 0$

0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0

$k = 1$

0.0	-1.0	-1.0	-1.0
-1.0	-1.0	-1.0	-1.0
-1.0	-1.0	-1.0	-1.0
-1.0	-1.0	-1.0	0.0

$k = 2$

	-1.7		
		-2.0	

- Two array updates using $v_{k=1}$ and v_k

$$v_{k+1}(s) \leftarrow \sum_a \pi(a|s) \sum_{s',r} p(s', r|s, a) \left[r + \gamma v_k(s')\right]$$

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Policy Evaluation in Gridworld

v_π for random policy

$k = 0$

0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0

$k = 1$

0.0	-1.0	-1.0	-1.0
-1.0	-1.0	-1.0	-1.0
-1.0	-1.0	-1.0	-1.0
-1.0	-1.0	-1.0	0.0

$k = 2$

0.0	-1.7	-2.0	-2.0
-1.7	-2.0	-2.0	-2.0
-2.0	-2.0	-2.0	-1.7
-2.0	-2.0	-1.7	0.0

- Two array updates using $v_{k=1}$ and v_k

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Policy Evaluation in Gridworld

v_π for random policy

$k = 3$

0.0	-2.4	-2.9	-3.0
-2.4	-2.9	-3.0	-2.9
-2.9	-3.0	-2.9	-2.4
-3.0	-2.9	-2.4	0.0

$k = 10$

0.0	-6.1	-8.4	-9.0
-6.1	-7.7	-8.4	-8.4
-8.4	-8.4	-7.7	-6.1
-9.0	-8.4	-6.1	0.0

$k = \infty$

0.0	-14	-20	-22
-14	-18	-20	-20
-20	-20	-18	-14
-22	-20	-14	0.0

- Two array updates using $v_{k=1}$ and v_k

$$v_{k+1}(s) \leftarrow \sum_a \pi(a|s) \sum_{s',r} p(s',r|s,a)\left[r + \gamma v_k(s')\right]$$

- Uniform random policy $\pi(a|s) = 0.25$ for all $a \in \mathcal{A} = \{\texttt{up}, \texttt{down}, \texttt{right}, \texttt{left}\}$ and for all non-terminal states $s \in \mathcal{S}$

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Policy Improvement Theorem

Let π and π' be any pair of deterministic policies such that, for all $s \in \mathcal{S}$,

$$q_\pi(s, \pi'(s)) \geq v_\pi(s)\,.$$

Then the policy π' must be as good as, or better than π.

- Recall π' is as good as, or better than π when

$$v_{\pi'}(s) \geq v_\pi(s) \text{ for all } s \in \mathcal{S}$$

How to Get the Improved Policy

- Old policy π, new policy π'
- We compute π' using the value function under policy π

$$
\begin{aligned}
\pi'(s) \leftarrow & \arg\max_a q_\pi(s,a) \\
& = \arg\max_a \mathbb{E}\left[R_{t+1} + \gamma v_\pi(S_{t+1}) \mid S_t = s, A_t = a\right] \\
& = \arg\max_a \sum_{s',r} p(s', r \mid s, a)\left[r + \gamma v_\pi(s')\right]
\end{aligned}
$$

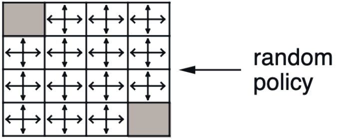
- Improved policy: take greedy action with respect to the value function of the original policy

Policy Improvement in Gridworld

v_k for the random policy | greedy policy w.r.t. v_k

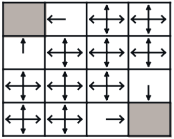
$k = 0$

0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0



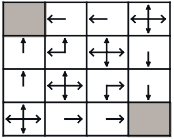
$k = 1$

0.0	-1.0	-1.0	-1.0
-1.0	-1.0	-1.0	-1.0
-1.0	-1.0	-1.0	-1.0
-1.0	-1.0	-1.0	0.0



$k = 2$

0.0	-1.7	-2.0	-2.0
-1.7	-2.0	-2.0	-2.0
-2.0	-2.0	-2.0	-1.7
-2.0	-2.0	-1.7	0.0



Policy Improvement in Gridworld

$k = 3$

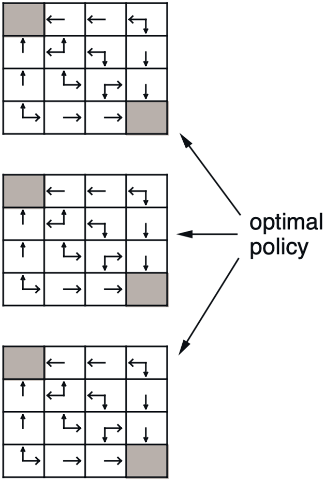
0.0	-2.4	-2.9	-3.0
-2.4	-2.9	-3.0	-2.9
-2.9	-3.0	-2.9	-2.4
-3.0	-2.9	-2.4	0.0

$k = 10$

0.0	-6.1	-8.4	-9.0
-6.1	-7.7	-8.4	-8.4
-8.4	-8.4	-7.7	-6.1
-9.0	-8.4	-6.1	0.0

$k = \infty$

0.0	-14.	-20.	-22.
-14.	-18.	-20.	-20.
-20.	-20.	-18.	-14.
-22.	-20.	-14.	0.0



Next class: Policy Iteration & Value Iteration (Chapter 4 of Sutton & Barto)